

Handheld Heart Rate Monitor

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Handheld devices play an essential role in the healthcare industry. With the sensors becoming smaller and compact, making a handheld device is also becoming easier and affordable. Heart rate measurement systems, or HR machines, are now available almost everywhere, as a feature in smartwatches to full-size stand-alone machines. Here is a simple handheld measurement system that can display a person's heart rate on an OLED display.

Connection diagram of the handheld heart rate monitor shown in Fig. 1 incorporates Arduino Nano (Board1), heart rate sensor, OLED display, and

3.7V Li-ion battery.

Heart rate sensor. It is used to capture the heart rate from a person's finger-tip and send as an analogue value to the microcontroller in Arduino board. The value, which ranges from 0 to 1024, is read on the Arduino's 10-bit analogue-to-digital (ADC) module. The value is then mapped to display the heart rate on OLED display.

Arduino Nano. Since the project aims to be cost-effective and efficient, Arduino Nano is a perfect controller for such an application. We want to have the freedom of having analogue pins and also a small form factor. The Arduino Nano also features an on-

board I2C data bus line, which makes the use of any I2C device possible and accessible over the controller. This controller has lower voltage requirements as well, making it perfect for battery based applications.

OLED display. It is a small 2.44cm

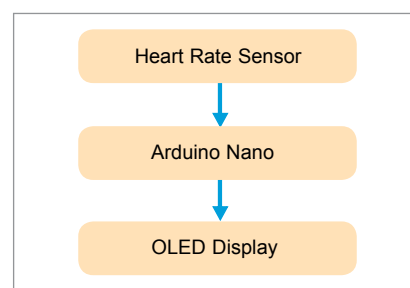


Fig. 2: Simplified flow chart of the program

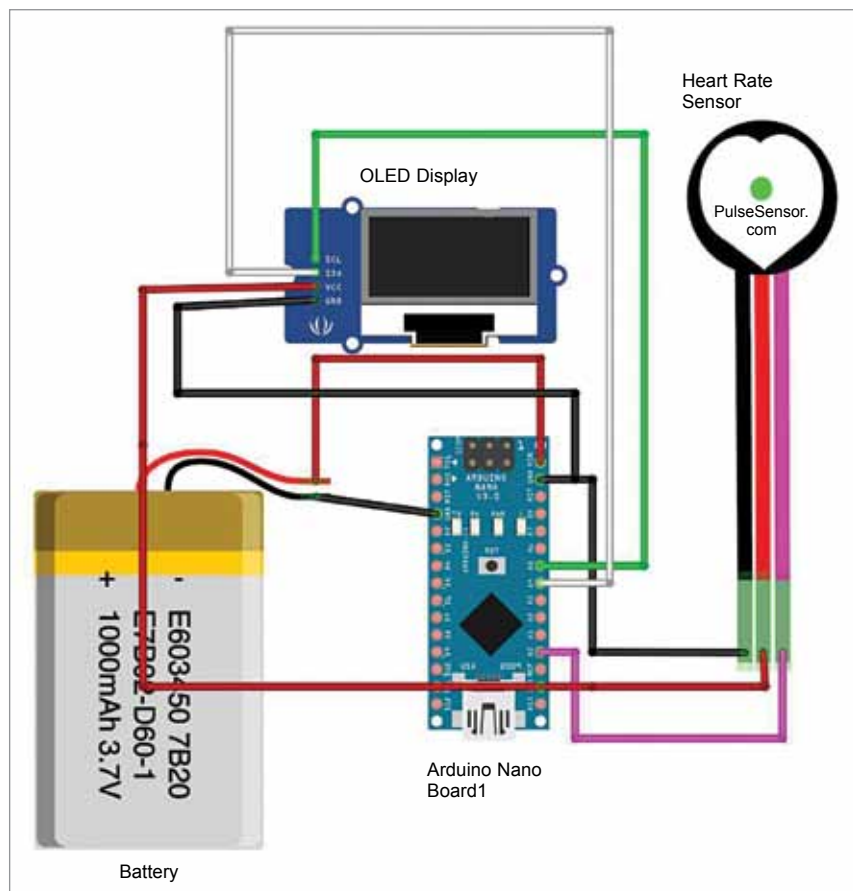


Fig. 1: Connection diagram

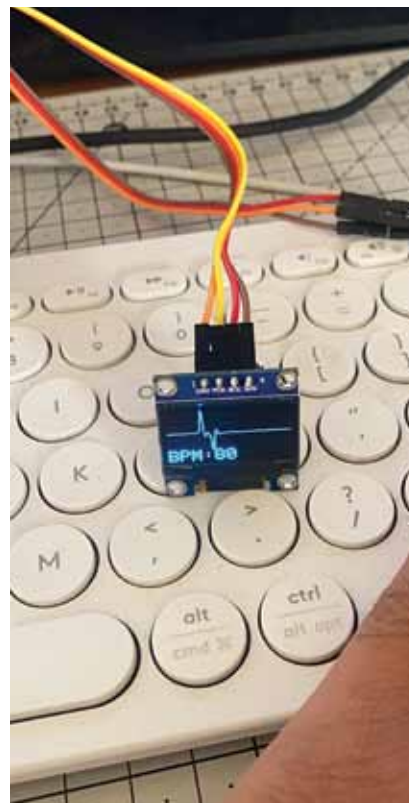


Fig. 3: The output of heart rate monitor with digital data and line graph